National Institute of Technology Patna

Department of Mathematics

End Semester Examination : December 2024

MA14102 : Engineering Mathematics I

Branch: CSE I (1st-Semester) **Maximum Marks:** 60

Time: 03.00 hours

Answer all questions

1. Find all the numbers x for which the vector $\begin{bmatrix} -3 \\ x \\ 13 \end{bmatrix}$ belongs to the subspace of $\mathbb{R}^3$ generated by the vectors $\begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ -4 \\ 9 \end{bmatrix}$. [6 Marks]

2. Find all the solutions of the system $2x + y + 2z = 3$, $3x - y + 4z = 7$ and $4x + 3y + 6z = 5$. Also find the null space associated with the coefficient matrix, its basis and dimension. [6 Marks]

3. Let $T : P_3[\mathbb{R}] \to P_2[\mathbb{R}]$ be defined by $T(a_0 + a_1t + a_2t^2 + a_3t^3) = a_1 + 2a_2t + 3a_3t^2$. Find the matrix representation of the linear transformation, $[T]_{B\to E}$ where $B = \{1, t, t^2, t^3\}$ and $E = \{1 + t, 1 - t, t^2\}$ are basis for $P_3[\mathbb{R}]$ and $P_2[\mathbb{R}]$, respectively. [6 Marks]

4. Find the non-singular matrix P that diagonalizes A and the diagonal matrix D such that $D = P^{-1}AP$ where [6 Marks]

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

5. If [6 Marks]

$$A = \begin{pmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{pmatrix}$$

verify Cayley-Hamilton theorem. Hence find A^{-1}.

6. By the method of variation of parameter solve [6 Marks]

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}.$$

7. (a) Solve $(1 + y^2)dx + (x - \tan^{-1} y)dy = 0$. [3 Marks]

(b) Solve the differential equation $ydx + (3 + 3x - y)dy = 0$ by making it exact. [3 Marks]

Please turn over...